



MINING DEALER BEST PRACTICE

Portable In-Pit Welding Shop for Shovel Bucket

**Maintenance and
Repair**

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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March 2013
1301-1.04-1301

1.0 Introduction

Each time a bucket intervention is required on a shovel's bucket, the mine needs to erect a portable shelter made of plywood around the bucket. It has been decided to have a long term vision for these frequent interventions. As the mine gets sub-zero temperatures 5 ½ months per year, provision needs to be made for shelter and heating for Workers. This shelter needs to be quickly commissioned to reduce downtime and gain efficiency.

2.0 Best Practice Description

A day or two prior to a service on the shovel, the portable welding shop is towed to a suitable site near the shovel. The compressor, generator and heating systems are hooked up along with the roll-off container containing the gases, welders and rods.

On the service day the roof and access doors are opened. The shovel walks a short distance to place its bucket on the 4 support legs and the shelter is then closed over the stick and bucket.

700000 BTU (5000 CFM) heating is made through holes in the floor that are connected through the under floor heat plenum. Three fan with 3000 CFM each are install at the top to evacuate welding fume and gas.

Powerful 12 volts LED lighting is permanently installed in the roof.

3.0 Implementation Steps

Axles and steel were purchased to fabricate the trailer and the sub-floor for the heating plenum. The building itself had to be dimensioned to accommodate the loading tool configuration. Different shaped wood panels for different shovels can be fabricated to wrap around the stick. The moveable roof section glides on rollers to open and close. The dimension for the welding shop is 26' large x28' long x16' high

4.0 Benefits

- 4-5 days time gains on every intervention compared to the former building of a wooden shed.
- Dramatic improvement on welder's efficiency.
- Work quality improvement linked to the more efficient set-up and elimination of wind drafts.
- Gains on employee's safety.

5.0 Resources Required

- Welder's team interviews to identify needs
- Engineer to draw the plans
- Welding team to build the structure
- Electricians
- Plumbers.

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6.0 Supporting Attachments / References



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Gooseneck to hook on tow truck

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7.0 Related Best Practices

8.0 Acknowledgements

Cliffs Natural Resources
BLOOM LAKE IRON ORE MINE
Fermont, Quebec, Canada

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